

The constant-Gaussian fluctuation model: Tonelli identity

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Abstract

First half of the `rem:pop_vs_emp` dichotomy (Astra #12 (b), #13 (c)). The leading coefficient is $A_p = \frac{y_{00}}{k_1 k_2} J_p(x_{00})$ with the positive Gaussian moment $J_p(x) = \int_0^\infty s^{p-1} e^{-\beta s^2} e^{\beta s x} ds$. For a Gaussian constant fluctuation $X \sim N(0, v)$ we prove, in the extended nonnegative reals,

$$\mathbb{E}_+[J_p(X)] = \int_0^\infty s^{p-1} e^{-\beta s^2} e^{v(\beta s)^2/2} ds$$

(`lintegral_gaussMomentJ_eq`), by Tonelli and the Gaussian moment generating function. The right-hand side is finite iff $\beta v < 2$ (next unit): the expected leading coefficient can be infinite although every sample coefficient is finite. Zero `sorry/axiom`.

1 The things formalised

```
def gaussMomentJ ( p x : ) :   := logMoment  p 0 (fun s => Real.exp ( * s * x))
theorem gaussMomentJ_pos (h) (hp) : 0 < gaussMomentJ  p x
theorem measurable_gaussMoment_integrand ( p) (X : Ω → ) (hX : Measurable X) :
  Measurable fun q : Ω ×   => ENNReal.ofReal (q.2^(p-1) * log q.2 ^ 0
    * (exp (- * q.2^2) * exp ( * q.2 * X q.1)))
theorem lintegral_exp_gaussian (X) (hX : Measurable X) (v : NNReal)
  (hXg : .map X = gaussianReal 0 v) (t) :
    , ENNReal.ofReal (exp (t * X ))   = ENNReal.ofReal (exp (v * t ^ 2 / 2))
theorem lintegral_gaussMomentJ_eq ( p) (h) (hp) (X) (hX) (v) (hXg) :
  , ENNReal.ofReal (gaussMomentJ  p (X ))
    = s in Ioi 0, ENNReal.ofReal (s^(p-1) * exp (- * s^2) * exp (v * ( * s)^2 / 2))
```

2 Mathematics

Each $J_p(X(\omega))$ is a Bochner integral of a positive integrable function, hence equal (after `ofReal`) to the corresponding Lebesgue integral. The joint integrand is measurable, so Tonelli swaps the order; the inner integral over ω is the exponential moment $\mathbb{E}[e^{\beta s X}] = e^{v(\beta s)^2/2}$ (Mathlib's `mgf_gaussianReal`, transported from the pushforward measure by `integrable_map_measure`). Isolating the positive moment J_p from the prefactor $y_{00}/(k_1 k_2)$, as Astra #12 advised, keeps the sign cases out of the core identity.

3 Paper-facing meaning

`rem:pop_vs_emp` states that the expected empirical coefficient differs from the population one because the coefficients depend nonlinearly on the fluctuation; the present identity quantifies this

for the simplest random fluctuation (a Gaussian constant): the expectation replaces $e^{-\beta s^2}$ by $e^{-(\beta-\beta^2 v/2)s^2}$, so $\mathbb{E}[J_p(X)] = J_p(0)(1 - \beta v/2)^{-p/2} > J_p(0)$ when $\beta v < 2$ and $= +\infty$ when $\beta v \geq 2$ (next unit). As Astra #12 stressed, this concerns the expected asymptotic coefficient, not the fixed- n chart integral on a compact chart, whose expectation stays finite.

4 Lean notes

- `integrable_map_measure hg hX.aemeasurable` transports integrability along `.map X = gaussianReal 0 v`; the a.e.-strong-measurability argument must be given as a separate `have` (an inline dot-chain mis-associates).
- `lintegral_lintegral_swap` needs `AEMeasurable (uncurry f)` on the product; `Measurable.pow_const` covers real powers of the coordinate.
- `ENNReal.ofReal_mul` (with the nonnegativity of the left factor) and `lintegral_const_mul` pull the s -dependent factor out of the ω -integral.